

UFO-sightings-Dashboard

1. What shapes of UFOs are observed most often?
2. Which US states report the most UFO sightings?
3. What are the annual trends in sightings since 2000?
4. How does the distribution of UFO sighting durations appear on the histogram (prior data cleaning will be needed)

```
In [3]: import pandas as pd
```

```
In [5]: df = pd.read_csv("ufo_sightings_scrubbed.csv")  
df.head()
```

C:\Users\Marcin\AppData\Local\Temp\ipykernel_19780\1683944241.py:1: DtypeWarning: Columns (5,9) have mixed types. Specify dtype option on import or set low_memory=False.

```
df = pd.read_csv("ufo_sightings_scrubbed.csv")
```

Out[5]:

	datetime	city	state	country	shape	duration (seconds)	duration (hours/min)	comments	date posted	latitude	longitude
0	1949-10-10 20:30:00	san marcos	tx	us	cylinder	2700	45 minutes	This event took place in early fall around 194...	2004-04-27	29.8830556	-97.941111
1	1949-10-10 21:00:00	lackland afb	tx	NaN	light	7200	1-2 hrs	1949 Lackland AFB, TX. Lights racing across...	2005-12-16	29.38421	-98.581082
2	1955-10-10 17:00:00	chester (uk/england)	NaN	gb	circle	20	20 seconds	Green/Orange circular disc over Chester, En...	2008-01-21	53.2	-2.916667
3	1956-10-10 21:00:00	edna	tx	us	circle	20	1/2 hour	My older brother and twin sister were leaving ...	2004-01-17	28.9783333	-96.645833
4	1960-10-10 20:00:00	kaneohe	hi	us	light	900	15 minutes	AS a Marine 1st Lt. flying an FJ4B fighter/att...	2004-01-22	21.4180556	-157.803611

In [7]: `df.shape`

Out[7]: (80332, 11)

In [9]: `df.dtypes`

```
Out[9]: datetime      object
        city          object
        state         object
        country       object
        shape         object
        duration (seconds) object
        duration (hours/min) object
        comments      object
        date posted   object
        latitude      object
        longitude     float64
        dtype: object
```

```
In [11]: df.isnull().sum()
```

```
Out[11]: datetime      0
        city          0
        state         5797
        country       9670
        shape         1932
        duration (seconds) 0
        duration (hours/min) 0
        comments      15
        date posted   0
        latitude      0
        longitude     0
        dtype: int64
```

1. What shapes of UFOs are observed most often?

```
In [13]: shape_counts = df["shape"].value_counts()
        print(shape_counts)
```

```
shape
light      16565
triangle   7865
circle     7608
fireball   6208
other      5649
unknown    5584
sphere     5387
disk       5213
oval       3733
formation  2457
cigar      2057
changing   1962
flash      1328
rectangle  1297
cylinder   1283
diamond    1178
chevron    952
egg        759
teardrop   750
cone       316
cross      233
delta      7
round      2
crescent   2
pyramid    1
flare      1
hexagon    1
dome       1
changed    1
Name: count, dtype: int64
```

```
In [15]: df["shape"] = df["shape"].fillna("unknown")
shape_counts = df["shape"].value_counts()
print(shape_counts)
```

shape	
light	16565
triangle	7865
circle	7608
unknown	7516
fireball	6208
other	5649
sphere	5387
disk	5213
oval	3733
formation	2457
cigar	2057
changing	1962
flash	1328
rectangle	1297
cylinder	1283
diamond	1178
chevron	952
egg	759
teardrop	750
cone	316
cross	233
delta	7
round	2
crescent	2
pyramid	1
flare	1
hexagon	1
dome	1
changed	1

Name: count, dtype: int64

```
In [17]: df["shape"] = df["shape"].fillna("unknown")
shape_percent = df["shape"].value_counts(normalize=True) * 100
shape_percent = shape_percent.round(2)
print(shape_percent)
```

```
shape
light      20.62
triangle   9.79
circle     9.47
unknown    9.36
fireball   7.73
other       7.03
sphere     6.71
disk        6.49
oval        4.65
formation  3.06
cigar       2.56
changing    2.44
flash       1.65
rectangle   1.61
cylinder    1.60
diamond     1.47
chevron     1.19
egg          0.94
teardrop    0.93
cone        0.39
cross       0.29
delta       0.01
round       0.00
crescent    0.00
pyramid     0.00
flare       0.00
hexagon     0.00
dome        0.00
changed     0.00
Name: proportion, dtype: float64
```

```
In [19]: df["shape"] = df["shape"].fillna("unknown")
shape_counts = df["shape"].value_counts()
shape_percent = df["shape"].value_counts(normalize=True) * 100
shape_summary = pd.DataFrame({
    "count": shape_counts,
    "percentage": shape_percent.round(2)
})
print(shape_summary)
```

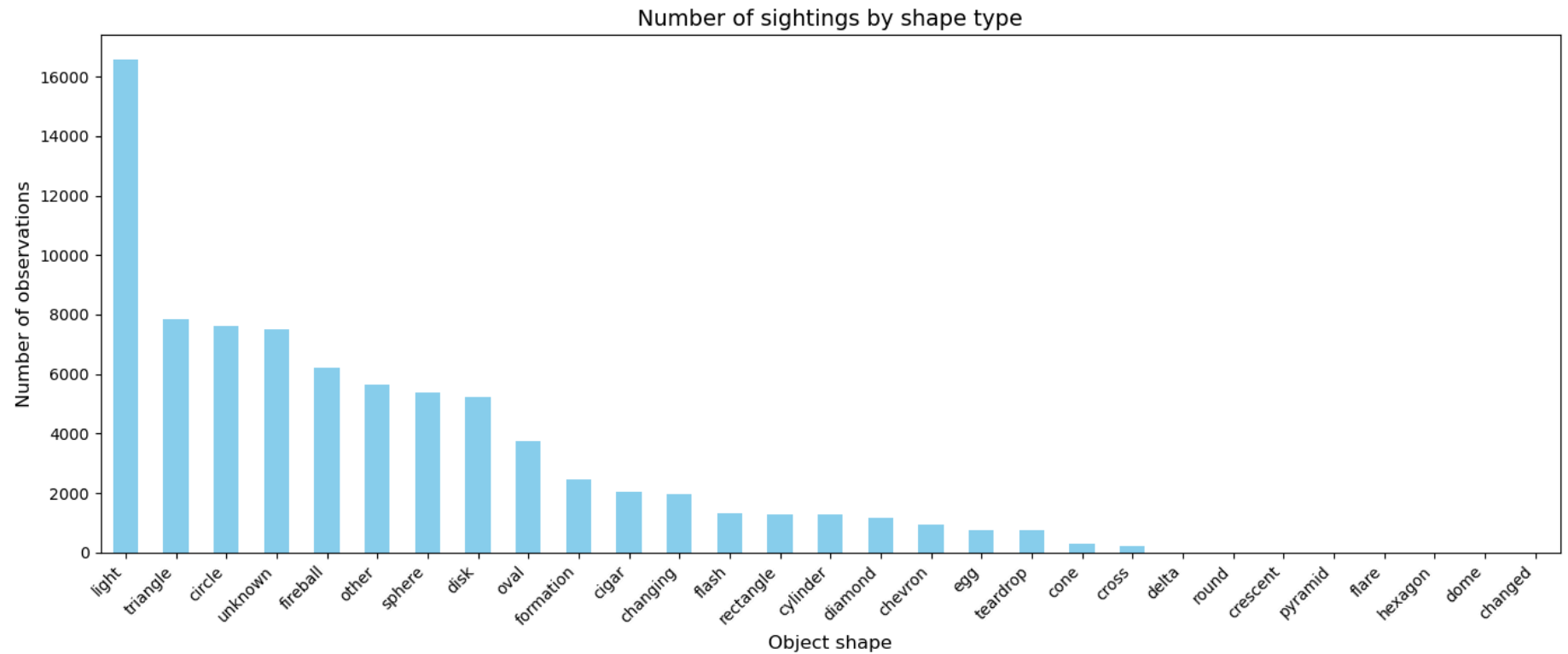
	count	percentage
shape		
light	16565	20.62
triangle	7865	9.79
circle	7608	9.47
unknown	7516	9.36
fireball	6208	7.73
other	5649	7.03
sphere	5387	6.71
disk	5213	6.49
oval	3733	4.65
formation	2457	3.06
cigar	2057	2.56
changing	1962	2.44
flash	1328	1.65
rectangle	1297	1.61
cylinder	1283	1.60
diamond	1178	1.47
chevron	952	1.19
egg	759	0.94
teardrop	750	0.93
cone	316	0.39
cross	233	0.29
delta	7	0.01
round	2	0.00
crescent	2	0.00
pyramid	1	0.00
flare	1	0.00
hexagon	1	0.00
dome	1	0.00
changed	1	0.00

```
In [21]: import matplotlib.pyplot as plt

plt.figure(figsize=(14, 6))
shape_counts.plot(kind='bar', color='skyblue')

plt.title('Number of sightings by shape type', fontsize=14)
plt.xlabel('Object shape', fontsize=12)
plt.ylabel('Number of observations', fontsize=12)
plt.xticks(rotation=45, ha='right')
```

```
plt.tight_layout()
plt.show()
```



2. Which US states report the most UFO sightings?

```
In [21]: country_counts = df["country"].value_counts()
print(country_counts)
```

```
country
us      65114
ca       3000
gb       1905
au        538
de        105
Name: count, dtype: int64
```



```
In [23]: df_us = df[df["country"] == "us"]  
state_counts = df_us["state"].value_counts()  
print(state_counts)
```

state

ca	8912
wa	3966
fl	3835
tx	3447
ny	2980
il	2499
az	2414
pa	2366
oh	2275
mi	1836
or	1747
nc	1740
mo	1458
co	1413
in	1288
va	1273
ma	1256
ga	1255
nj	1255
wi	1232
tn	1119
mn	1012
sc	1003
ct	892
ky	855
md	837
nv	803
ok	724
nm	720
ia	678
al	642
ut	622
ks	613
ar	588
la	558
me	558
id	521
nh	486
mt	478
wv	448

```
ne    381
ms    375
ak    319
hi    262
vt    260
ri    228
sd    183
wy    175
de    166
nd    129
pr     25
dc      7
Name: count, dtype: int64
```

```
In [25]: df_us = df[df["country"] == "us"]
state_percent = df_us["state"].value_counts(normalize=True) * 100
state_percent = state_percent.round(2)
print(state_percent)
```

state	
ca	13.69
wa	6.09
fl	5.89
tx	5.29
ny	4.58
il	3.84
az	3.71
pa	3.63
oh	3.49
mi	2.82
or	2.68
nc	2.67
mo	2.24
co	2.17
in	1.98
va	1.96
ma	1.93
ga	1.93
nj	1.93
wi	1.89
tn	1.72
mn	1.55
sc	1.54
ct	1.37
ky	1.31
md	1.29
nv	1.23
ok	1.11
nm	1.11
ia	1.04
al	0.99
ut	0.96
ks	0.94
ar	0.90
la	0.86
me	0.86
id	0.80
nh	0.75
mt	0.73
wv	0.69

```
ne    0.59
ms    0.58
ak    0.49
hi    0.40
vt    0.40
ri    0.35
sd    0.28
wy    0.27
de    0.25
nd    0.20
pr    0.04
dc    0.01
```

Name: proportion, dtype: float64

```
In [27]: df_us = df[df["country"] == "us"]
state_counts = df_us["state"].value_counts()
state_percent = df_us["state"].value_counts(normalize=True) * 100
state_summary = pd.DataFrame({
    "count": state_counts,
    "percentage": state_percent.round(2)
})
print(state_summary)
```

state	count	percentage
ca	8912	13.69
wa	3966	6.09
fl	3835	5.89
tx	3447	5.29
ny	2980	4.58
il	2499	3.84
az	2414	3.71
pa	2366	3.63
oh	2275	3.49
mi	1836	2.82
or	1747	2.68
nc	1740	2.67
mo	1458	2.24
co	1413	2.17
in	1288	1.98
va	1273	1.96
ma	1256	1.93
ga	1255	1.93
nj	1255	1.93
wi	1232	1.89
tn	1119	1.72
mn	1012	1.55
sc	1003	1.54
ct	892	1.37
ky	855	1.31
md	837	1.29
nv	803	1.23
ok	724	1.11
nm	720	1.11
ia	678	1.04
al	642	0.99
ut	622	0.96
ks	613	0.94
ar	588	0.90
la	558	0.86
me	558	0.86
id	521	0.80
nh	486	0.75
mt	478	0.73

wv	448	0.69
ne	381	0.59
ms	375	0.58
ak	319	0.49
hi	262	0.40
vt	260	0.40
ri	228	0.35
sd	183	0.28
wy	175	0.27
de	166	0.25
nd	129	0.20
pr	25	0.04
dc	7	0.01

3. What are the annual trends in sightings since 2000?

```
In [29]: df["datetime"] = pd.to_datetime(df["datetime"], errors='coerce')
df["year"] = df["datetime"].dt.year
df.head()
```

Out[29]:

	datetime	city	state	country	shape	duration (seconds)	duration (hours/min)	comments	date posted	latitude	longitude	year
0	1949-10-10 20:30:00	san marcos	tx	us	cylinder	2700	45 minutes	This event took place in early fall around 194...	2004-04-27	29.8830556	-97.941111	1949
1	1949-10-10 21:00:00	lackland afb	tx	NaN	light	7200	1-2 hrs	1949 Lackland AFB, TX. Lights racing across...	2005-12-16	29.38421	-98.581082	1949
2	1955-10-10 17:00:00	chester (uk/england)	NaN	gb	circle	20	20 seconds	Green/Orange circular disc over Chester, En...	2008-01-21	53.2	-2.916667	1955
3	1956-10-10 21:00:00	edna	tx	us	circle	20	1/2 hour	My older brother and twin sister were leaving ...	2004-01-17	28.9783333	-96.645833	1956
4	1960-10-10 20:00:00	kaneohe	hi	us	light	900	15 minutes	AS a Marine 1st Lt. flying an FJ4B fighter/att...	2004-01-22	21.4180556	-157.803611	1960

In [31]:

```
df_filtered = df[df['year'] >= 2000]
year_counts = df_filtered["year"].value_counts().sort_index()
print(year_counts)
```



```
year
2000    2772
2001    3122
2002    3235
2003    3962
2004    4257
2005    4083
2006    3721
2007    4269
2008    4819
2009    4542
2010    4283
2011    5107
2012    7356
2013    7038
2014    2260
Name: count, dtype: int64
```

```
In [80]: import seaborn as sns
```

```
df['datetime'] = pd.to_datetime(df['datetime'], errors='coerce')
df = df.dropna(subset=['datetime'])
df['year'] = df['datetime'].dt.year

df_filtered = df[df['year'] >= 2000]

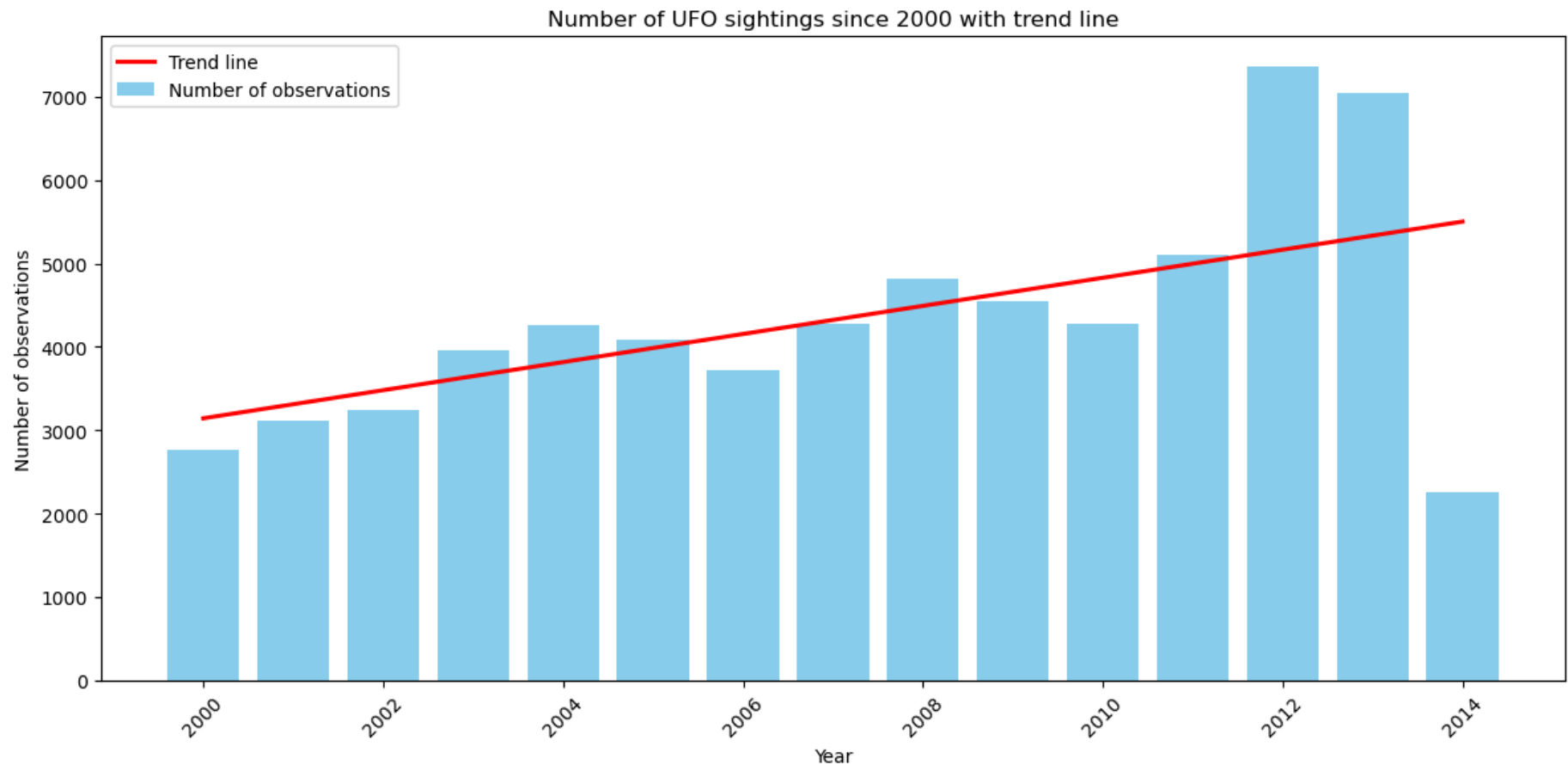
year_counts = df_filtered['year'].value_counts().sort_index()
year_df = year_counts.reset_index()
year_df.columns = ['year', 'count']
year_df['year'] = year_df['year'].astype(int)

plt.figure(figsize=(12, 6))
plt.bar(year_df['year'], year_df['count'], color='skyblue', label='Number of observations')

sns.regplot(x=year_df['year'], y=year_df['count'], scatter=False, color='red', label='Trend line', ci=None)

plt.title('Number of UFO sightings since 2000 with trend line')
plt.xlabel('Year')
plt.ylabel('Number of observations')
plt.xticks(rotation=45)
plt.legend()
```

```
plt.tight_layout()
plt.show()
```



4. How does the distribution of UFO sighting durations appear on the histogram (prior data cleaning will be needed)

```
In [55]: df['duration (seconds)'] = pd.to_numeric(df['duration (seconds)'], errors='coerce')

bins = [0, 59, 299, 599, 1199, 1799, 3600, float('inf')]
labels = [
    '< 1 min',
    '1-5 min',

```

```

    '5-10 min',
    '10-20 min',
    '20-30 min',
    '30-60 min',
    '> 60 min'
]
df['duration_group'] = pd.cut(df['duration (seconds)'], bins=bins, labels=labels, right=True)

duration_counts = df['duration_group'].value_counts().sort_index()
print(duration_counts)

```

```

duration_group
< 1 min      23584
1-5 min      21210
5-10 min     10882
10-20 min    10976
20-30 min     3674
30-60 min     6802
> 60 min     3201
Name: count, dtype: int64

```

```

In [82]: duration_counts = df['duration_group'].value_counts().sort_index()
duration_percent = (duration_counts / duration_counts.sum()) * 100
duration_summary = pd.DataFrame({
    'Number of observations': duration_counts,
    'Percentage of sightings [%]': duration_percent.round(2)
})
print(duration_summary)

```

duration_group	Number of observations	Percentage of sightings [%]
< 1 min	23584	29.36
1-5 min	21210	26.40
5-10 min	10882	13.55
10-20 min	10976	13.66
20-30 min	3674	4.57
30-60 min	6802	8.47
> 60 min	3201	3.98

```

In [84]: duration_counts = df['duration_group'].value_counts().sort_values(ascending=False)

```

```
cumulative_percentage = duration_counts.cumsum() / duration_counts.sum() * 100

fig, ax1 = plt.subplots(figsize=(10, 6))

ax1.bar(duration_counts.index.astype(str), duration_counts.values, color='skyblue')
ax1.set_xlabel('Groups by duration of sightings')
ax1.set_ylabel('Number of observations', color='blue')
ax1.tick_params('y', colors='blue')
plt.xticks(rotation=45)

ax2 = ax1.twinx()
ax2.plot(duration_counts.index.astype(str), cumulative_percentage, color='red', marker='o', linestyle='-')
ax2.set_ylabel('Cumulative %', color='red')
ax2.tick_params('y', colors='red')
ax2.axhline(y=55, color='gray', linestyle='--', linewidth=1) # Line 55%

plt.title('Pareto chart for observation duration')
plt.tight_layout()
plt.show()
```

